DRAFT — P1–P5 MEASURED; P4-a-block-energy HIT (excess_final=0.08727119221832025); P5 2HIT/4MISS; L-cool-dir INSUFFICIENT

A3 Predictive Intuition Guide

Primary learning resource for Assignment 3 prediction mastery. Organized by handout problem (P1–P5), with a quantitative appendix merged in so this is the one comprehensive study PDF.

PDF / doc policy (max 2–3 final PDFs)

#	Final PDF candidate	Source in store
1	Assignment report (≤20 pages)	`docs/a3-assignment-report.md` (+ appendix MD, not a separate study PDF)
2	**This guide** (comprehensive)	`docs/a3-predictive-intuition-guide.md` — includes Appendix Q
3	Optional practice quiz + key	`docs/a3-counterfactual-prediction-quiz.md`

Not PDFs: prediction ledger (`docs/a3-prediction-ledger.md` + `.csv`), checklist (`internal/a3-requirements-checklist.md`), project context, experimental appendix tables, compact quant card `docs/a3-quantitative-prediction-reference.md` (MD only — mirrors Appendix Q MEASURED/HIT tables). Do not export overlapping study PDFs.

Do not invent measured numbers. Section **A** stays **PENDING MEASUREMENT** for any quantity not yet observed; when a stage lands, label those facts **MEASURED** and leave the rest **PENDING**. Sections **B–G** are mechanism hypotheses grounded in the handout (`internal/a3_optimization2.txt`). Pre-registered numerical intervals are labeled **PREDICTION** here and mirrored in the ledger (MD/CSV).

A–G schema (every subproblem)

Letter	Name	Fill rule
A	Experimental result	**PENDING MEASUREMENT** — placeholders only; never invent values
B	Landscape / geometry	Minima, basins, barriers, spectrum shape; equations from handout
C	Dynamics / optimizer interaction	Schedules, momentum, Adam preconditioning, bouncing
D	Quantitative prediction template	Intervals, rankings, thresholds tagged **PREDICTION**
E	Counterfactual levers	Single knobs that should flip the outcome
F	Failure modes / confounds	Bad LR, short horizon, probe noise, seed
G	Cross-links & further tests	Ties to other problems; next cheap check

Default setup unless overridden: d8 (~35M), ~614M tokens, AdamW (\beta_1,\beta_2=(0.9,0.95)), (\epsilon=10^{-8}), context 1024, batch 64, WD 0.1 (masked), grad clip 1.0; val loss ↓ better.

Problem 1 — Mode Connectivity

Handout probe: for convex (F), Jensen gives (F((x+y)/2)\le (F(x)+F(y))/2). Deep nets are nonconvex; midpoint barrier (B) tests whether two low-loss checkpoints share a basin in **parameter** space.

P1(a) — Forked-run sweep; endpoints + prediction ensemble

A. Experimental result — **MEASURED** (Batch 1 measure complete)

- **MEASURED (prefix):** run `model-d8-lr0.003-tok614M-wsd0.0-stop1000` (W&B `rq1k0bz1`); end `train_loss=3.43004`, `val_loss=3.48832`. Modal `ap-L6RKrCcPCbhhgBvr3DLJfU`.
- **MEASURED (branches):** 12/12 keep-ckpts; Modal `ap-57b3QORBZcKneFVkuuq7F3`.
- **MEASURED (30-cell grid):** raw JSON `media/p1/a3-mode-connectivity.json` (repo copy under `experiments/a3_optimization/results/`, 20091 B @ 2026-10-10 17:21 PDT); Modal measure app `ap-ZU7fsZIl4Mq2Wi5veNzXCb`. Full table: `docs/a3-experimental-appendix.md` / `media/p1/barrier-table.csv`.
- **MEASURED endpoints / ensemble (examples):** (0,500)(0,500): L1=3.7511L_1=3.7511, L2=3.7483L_2=3.7483, Lens=3.6454L_{\mathrm{ens}}=3.6454; (1000,25)(1000,25): L1=3.4786L_1=3.4786, L2=3.4796L_2=3.4796, Lens=3.4221L_{\mathrm{ens}}=3.4221; (250,100)(250,100): Lens=3.8869L_{\mathrm{ens}}=3.8869. When B>0.01B>0.01, ensemble is substantially below weight-avg (e.g. (0,500)(0,500): Lens=3.6454L_{\mathrm{ens}}=3.6454 vs L(θavg)=4.3734L(\theta_{\mathrm{avg}})=4.3734).

B. Landscape / geometry
For next-token probabilities, [p_{\mathrm{ens}}(\cdot\mid c)=\tfrac12 p_{\theta_1}(\cdot\mid c) +\tfrac12 p_{\theta_2}(\cdot\mid c).] Ensemble averaging is in **function space**. Two networks can disagree in (\theta) (high barrier later in P1b) yet agree on soft predictions → ensemble can stay low-loss while (\theta_{\mathrm{avg}}) sits on a ridge. Shared long prefix (x) keeps branches in a common basin; early forks ((x=0)) maximize opportunity for mode separation.

C. Dynamics / optimizer interaction
Required schedule: `wsd0.0` (constant after warmup). Fork copies **model + optimizer state** at step (x); branches use different `batch_order_seed` but the same data distribution. Constant LR keeps late-training bounce amplitude high (river-valley; P2), so stochastic data-order differences can accumulate into distinct wall-orbits even when final losses match.
FITTED (Batch 1): at early forks the ensemble stays low while (B) grows with post-fork horizon (y) (Appendix Q1.1 Fit A); shared-prefix length (x) collapses that barrier exponentially at fixed (y=500) (Fit B). Plots: `media/p1/plots/B_vs_y_curves.png`, `ensemble_vs_avg.png`.

D. Quantitative prediction template — calibrate against MEASURED where resolved

- **PREDICTION (ranking; still open):** for fixed (y), as (x) increases, (L(p_{\mathrm{ens}})) approaches the better endpoint (shared-basin soft agreement).
- **Ledger `P1-grid`:** PREDICTION was fraction (B\le 0.01) in ([0.2,0.5)). **MEASURED:** (26/30=0.867\in[0.8,1]) — prediction under-estimated connectivity (still concentrated: failures only at small (x), large (y)).
- **Ordering:** where modes separate, (L(p_{\mathrm{ens}}) < L(\theta_{\mathrm{avg}})). **MEASURED** on all four (B>0.01) cells (gaps below in P1b / ledger `P1-ens-gap`).

E. Counterfactual levers
Change only one of: (x), (y), `batch_order_seed` pair, peak LR (keep schedule `wsd0.0` for the required grid). Optional later: same cells under weight averaging (P1b) or decay schedules (P1d).

F. Failure modes / confounds
Untuned peak LR making both endpoints bad; comparing cells with unequal total steps (x+y); evaluating before branches finish (y) updates; mixing train vs val loss when reporting endpoints.

G. Cross-links & further tests
Ensemble softness foreshadows P2 EMA (parameter-space averaging along a trajectory). Next cheap check: same cells under (\theta_{\mathrm{avg}}) and (B) (P1b). Cross-read: P4 (\lambda_{\max}) may differ across basins even when (L) matches.

P1(b) — Weight averaging and midpoint barrier (B)

A. Experimental result — MEASURED

- Definition used: B=L(θavg)−12(L1+L2)B=L(\theta_{\mathrm{avg}})-\tfrac12(L_1+L_2) at midpoint α=0.5\alpha=0.5 from measure JSON.
- **MEASURED:** Bmin=−0.1161B_{\min}=-0.1161 at (250,100)(250,100); Bmax=0.6237B_{\max}=0.6237 at (0,500)(0,500); mean −0.0334-0.0334.
- **MEASURED: 26 / 30** cells have B≤0.01B\le 0.01. Failures (B>0.01B>0.01): (0,250)(0,250) B=0.2173B=0.2173, (0,500)(0,500) B=0.6237B=0.6237, (50,500)(50,500) B=0.1299B=0.1299, (100,500)(100,500) B=0.0319B=0.0319.
- **MEASURED pattern:** large BB concentrated at small xx and large yy; all x≥250x\ge 250 cells (and all x=1000x=1000) satisfy B≤0.01B\le 0.01. Table: `media/p1/barrier-table.csv`.

B. Landscape / geometry
[B(\theta_1,\theta_2)=L!\left(\tfrac{\theta_1+\theta_2}{2}\right)-\tfrac{L(\theta_1)+L(\theta_2)}{2},\qquad \theta_{\mathrm{avg}}=\tfrac12\theta_1+\tfrac12\theta_2.]
Handout criterion: (B\le 0.01) ⇔ “weight averaging works.” Large (B) ⇒ linear path crosses a ridge between modes. (B) is a **midpoint** probe only — full curves in P1(c).

C. Dynamics / optimizer interaction
High constant LR encourages bouncing on valley walls (Wen et al. river-valley image in handout).

Branches that bounce differently accumulate parameter disagreement ($|\theta_1-\theta_2|$) even if ($L(\theta_1)\approx L(\theta_2)$). Longer shared prefix synchronizes both weights and Adam moments before the fork.

FITTED rates (MEASURED grid only):

- At (x=0), growth regime ($y\ge 100$): $B\approx -0.190+0.00163,y$ ($R^2\approx 1.00$), (n=3)) — barrier rises ~0.16 per +100 post-fork steps.
- At (y=500): $B\approx -0.066+0.687,e^{-0.0232,x}$ ($R^2=0.995$)) — shared prefix halves the excess barrier on a ~30-step (e)-folding.
- At ($x\ge 500$), slope of (B) vs (y) is consistent with zero ($R^2\le 0.26$)) — late forks stay connected across the measured (y) range.
  Heatmap: `media/p1/plots/B_heatmap.png`. Full equations / residuals: Appendix Q1.1.

D. Quantitative prediction template — PREDICTION vs MEASURED vs FITTED

- Threshold map intuition: large (x), small (y) → ($B\le 0.01$); small (x), large (y) → ($B>0.01$).
  MEASURED: matches direction (all failures have ($x\le 100$) and ($y\ge 250$)).
- Ledger `P1-B-early-long`:
  - **PREDICTION:** $B(0,500)\in[0.05,0.2)$. **MEASURED:** $B(0,500)=0.6237\in[0.2,\infty)$ — higher than predicted.
  - **PREDICTION:** $B(1000,25)\in(-\infty,0.01]$. **MEASURED:** $B(1000,25)=-0.0597$ — match.
- Ledger `P1-ens-gap` (cells with ($B>0.01$)): let $G=L(\theta_{\mathrm{avg}})-L(p_{\mathrm{ens}})$.
  - **PREDICTION:** ($G\ge 0.01$) and median ($G\in[0.02,0.08)$).
  - **MEASURED gaps:** ((100,500),0.1311); ((50,500),0.2309); ((0,250),0.3273); ((0,500),0.7281); median ($G\approx 0.279\in[0.2,\infty)$). Direction ($G\ge 0.01$)) matched; magnitude under-predicted.
- **FITTED:** $G\approx 0.090+1.033,B$ on all 30 cells ($R^2=0.989$)); on ($B>0.01$) cells $G\approx 0.102+1.007,B$ ($R^2=0.9997$)).
- **PREDICTION (extrapolated; ledger):** `P1-B-x0-y750`, `P1-B-x25-y500`, `P1-B-x2000-y500`, `P1-G-x0-y750` — intervals from Fits A–C beyond the measured ((x,y)) grid (Appendix Q1.1).

E. Counterfactual levers

Peak LR; batch size; schedule after fork (P1d); measuring only midpoints vs full (α) curves (P1c).

F. Failure modes / confounds

Barrier criterion is on **validation** loss — train-loss barriers can differ. Checkpoint timing must match (x+y). Cherry-picking only low-(B) cells for narrative. Treating ($B\le 0.01$) as continuous quality rather than a binary handout threshold.

G. Cross-links & further tests

Where (B) is large, expect ensembles ≪ weight avg (P1a). Ties to P4: different basins may show different (\lambda_{\max}) under the same loss band. P2: decaying LR should shrink typical (B) at fixed ((x,y)) (P1d).

P1(c) — Full interpolation curves (\theta(\alpha))

A. Experimental result — **MEASURED** (5-point α\alpha curves in JSON)

- Every cell has val losses at α∈{0,0.25,0.5,0.75,1}\alpha\in\{0,0.25,0.5,0.75,1\} in `media/p1/a3-mode-connectivity.json`.
- **MEASURED examples:** (0,500)(0,500) interpolation `[3.7511, 3.9156, 4.3734, 3.8961, 3.7483]` with Lens=3.6454L_{\mathrm{ens}}=3.6454 (large midpoint barrier). (1000,500)(1000,500): `[3.3812, 3.317, 3.2986, 3.3172, 3.381]` with Lens=3.3017L_{\mathrm{ens}}=3.3017, B=−0.0825B=-0.0825.
- Full per-cell curves: raw JSON; midpoint summary: appendix / `barrier-table.csv`.

B. Landscape / geometry
[\theta(\alpha)=(1-\alpha)\theta_1+\alpha\theta_2,\quad \alpha\in{0,0.25,0.5,0.75,1}.]
Unimodal barrier peaking near (\alpha=0.5) ⇒ distinct basins along the linear path. Near-flat or concave path ⇒ mode connectivity / shared basin. Asymmetric peak ((\alpha\neq 0.5)) means midpoint (B) understates the true path max.

C. Dynamics / optimizer interaction
Endpoints from the same constant-LR fork often have similar (|\theta|) scale, so linear interpolation is a meaningful geometric probe. Midpoint alone can miss asymmetric barriers; denser (\alpha) later if asymmetry appears.
FITTED link: when midpoint (B) is large, Fit C gives (G\approx B+0.10), so the (\alpha=0.5) ridge height almost fully appears as ensemble–weight-avg gap (see `media/p1/plots/ensemble_vs_avg.png`).

D. Quantitative prediction template — PREDICTION vs MEASURED (ledger `P1-interp-peak`)
For a representative cell with midpoint (B\in[0.05,0.2)) — **MEASURED** cell ((50,500)), (B=0.1299):

- **PREDICTION (ranking):** (L(\theta(0.5))) strictly largest among five (\alpha).
  MEASURED: ([3.7009, 3.7345, 3.8253, 3.7265, 3.6898]) — (\alpha=0.5) worst; match.
- **PREDICTION (height):** barrier in ([0.05,0.2)). **MEASURED:** (0.1299) — match.
- **PREDICTION:** (L(p_{\mathrm{ens}})) below barrier peak. **MEASURED:** (L_{\mathrm{ens}}=3.5943 < 3.8253=L(\theta(0.5))).
  High-(B) extreme ((0,500)): peak still at (\alpha=0.5), but height (0.6237\in[0.2,\infty)) (outside the preregistered mid band).

E. Counterfactual levers
Which ((x,y)) cells are “representative”; optional denser (\alpha) grid; compare a low-(B) cell (flat path) vs high-(B) cell (barrier).

F. Failure modes / confounds
Cherry-picking only low-(B) cells; reporting train loss on the path; numerical issues if endpoints have wildly different scales (unlikely same prefix).

G. Cross-links & further tests
Use curves to choose P1(d) ablations. Connect to P2: EMA averages along a **trajectory**, not across independently forked modes — geometry differs from (\theta(\alpha)).

P1(d) — What controls same-basin membership

A. Experimental result — PENDING MEASUREMENT

- TODO: LR / batch / schedule ablations at fixed fork geometry; patterns; proposed follow-ups.

B. Landscape / geometry
Basin identity is not final loss alone — two low-loss solutions can sit across a ridge ((B>0.01) with matched endpoints). Knobs that keep trajectories closer in parameter space raise the chance of (B\le 0.01).

C. Dynamics / optimizer interaction

- Lower peak LR or decay schedules → less wall-bounce → branches stay closer.
- Larger batch → less gradient noise → slower stochastic mode separation.
- `wsd0.0` after fork maximizes continued bounce; cosine / `wsd0.2` settle toward the river late (P2).
  A2 lesson: retune peak LR after batch changes or conclusions are LR artifacts.
  FITTED baseline (constant `wsd0.0` only): Fit B says excess barrier at (y=500) decays as (\sim e^{-0.023,x}); schedule ablations should be scored as (\Delta B) vs this constant-LR map (Appendix Q1.1), not vs a blank slate.

D. Quantitative prediction template — PREDICTION (ledger `P1-d-dB-cos`)
Fix ((x,y)=(250,250)), same seeds, same peak LR; arm W = `wsd0.0` after fork; arm C = cosine after fork; (\Delta B=B_W-B_C).

- **PREDICTION:** (\Delta B\in[0.02,0.1)) (cosine lowers barrier).
- **PREDICTION:** (B_C\le 0.01) only if (B_W) was modestly above 0.01 — else cosine helps but may not cross the threshold (forced choice: “depends on (B_W) band”).
  Open pre-reg (ledger notes): peak-LR (\times 2) and batch (\times 2) (\Delta B) intervals before launch.

E. Counterfactual levers
Peak LR (\times 2); batch (\times 2); `wsd0.2` vs `wsd0.0`; seed diversity; warmup length (secondary).

F. Failure modes / confounds
Changing multiple knobs at once; not retuning LR after batch change; claiming “schedule causes basins” without matched token budgets / evaluation protocol.

G. Cross-links & further tests
P2 regret/EMA: same bounce mechanism. P3: edge-of-stability regime ((\eta\lambda) large) may correlate with high (B) under constant LR. Cheap next: one cosine vs `wsd0.0` cell before a full grid.

Problem 2 — River-Valley Dynamics

Handout picture (Wen et al.): steep hillsides + flat river. Large step sizes bounce between walls while drifting along the river; decay settles iterates toward the river.

Batch 2 label snapshot (finished 9/9 required + P3 sharpness — do not invent the rest):

Status	Items
MEASURED	Four schedule finals; dip **final** (R); invsqrt (\Delta L); **SGDR** (\Delta L) + vs-invsqrt; P3 SGD (\lambda_{\max}); AdamW (\eta\lambda^{\mathrm{pre}}); **P4(a)** stage-a **13/13** scored; **P5** 6/7 (`P5-L-cool-dir` **INSUFFICIENT**)
PENDING (not MEASURED)	**mid-dip** (R_{\mathrm{mid}}); EMA gap; P3 η-change arms; **P4(b)** after-arm / `P4-invar-dL`; **P4(c)** (P4(b) soft baseline MEASURED — see §P4(b))

P2(a) — Schedule comparison (constant / linear / cosine / plateau-then-decay)

A. Experimental result — **MEASURED** (four schedule arms complete; Batch 2 overall **9/9** + P3 sharpness)

- **MEASURED finals** (`media/p2/batch2_partial_finals.csv`):

Rank	Arm	W&B	Final val	(R) vs best=`linear`
1	`linear`	`r0294p93`	2.9282028675079346	0
2	`cos`	`sktw67q8`	2.934119462966919	0.005916595458984375
3	`wsd0.2`	`soosjcxr`	2.93774151802063	0.009538650512695312
4	`wsd0.0`	`widt420l`	3.0975050926208496	0.16930222511291504

- **After measurement (ledger `B2-P2-*-R` / `P2-sched-rank`; scorecard):**
  - **Ranking HIT** (`P2-sched-rank`): (\min(L_{\mathrm{cos}},L_{\mathrm{wsd0.2}},L_{\mathrm{linear}}) < L_{\mathrm{wsd0.0}}) — direction of "decayed beats constant" matched.
  - **`wsd0.0` (R) band MISS:** MEASURED (R=0.16930222511291504\approx 0.17) vs PREDICTION ([0.02,0.08)); still worst of four (worst **HIT**). Constant LR regret was under-estimated.
  - **`wsd0.2` top-2 MISS:** band (R\in[0,0.02)) **HIT**, but rank #3 (top-2 with `cos` **MISS**); `linear`/`cos` took the top-2.
  - **`linear` / `cos`:** both band **HIT**; `linear` best ((R=0)).
- **Oct 20 quiz implication:** when scoring constant-LR vs decayed regret, prefer **higher** (R) bands than the locked `[0.02,0.08)` — the miss signal is that constant LR regret was under-

estimated (here (\approx 0.17)). Keep ranking direction (“decayed (<) constant”) unless a new measure flips it.
- Batch 2 **9/9**: dip final (R), invsqrt (\Delta L), SGDR, and both P3 sharpness arms also MEASURED (see P2(c)/(d) + P3 §A). Still **PENDING**: EMA / mid-dip.
- TODO still: LR/train/val curves; peak-LR robustness; EMA gap for P2(b).

B. Landscape / geometry
Constant late LR keeps large bounce amplitude → iterates spend time up the hillsides → higher final val possible even if mid-training loss looks competitive. Decay schedules shrink hillside steps → settle nearer the river floor.

C. Dynamics / optimizer interaction
Starter schedules: `wsd0.0` (constant after warmup), `linear`, `cos`, `wsd0.2` (decay last 20%). Compare at equal token counts. AdamW unchanged. Peak LR must be tuned enough that the ranking is not an under-tuned artifact (handout).

D. Quantitative prediction template — locked PREDICTION vs **MEASURED** (Batch 2 lock `2026-10-11T00:32:07Z`)

Ledger ID	Batch 2 job	Quantity	PREDICTION band	Status
`B2-P2-wsd0.0-R`	`a3-b2-p2-wsd0.0`	Final (R) vs best of four	(R\in[0.02,0.08)); worst	**MEASURED** (R=0.1693) band **MISS**; worst **HIT**
`B2-P2-linear-R`	`a3-b2-p2-linear`	Final (R)	(R\in[0,0.02)); (<L_{\mathrm{wsd0.0}})	**MEASURED HIT** ((R=0))
`B2-P2-cos-R`	`a3-b2-p2-cos`	Final (R)	(R\in[0,0.02)); top-2	**MEASURED HIT**
`B2-P2-wsd0.2-R`	`a3-b2-p2-wsd0.2`	Final (R)	(R\in[0,0.02)); top-2	**MEASURED** band **HIT**; top-2 **MISS**
`P2-sched-rank`	jobs 0–3	Ranking	(\min(L_{\mathrm{cos}},L_{\mathrm{wsd0.2}},L_{\mathrm{linear}}) < L_{\mathrm{wsd0.0}})	**MEASURED HIT**

Best raw final → (\theta^{\mathrm{ref}}) for (R_M(t)). Cross-refs: dip / HF in P2(c)/(d); P3 sharpness in P3 §A. Still **PENDING**: mid-dip (R_{\mathrm{mid}}), EMA.

E. Counterfactual levers
Peak LR; decay fraction (`wsd0.2` vs other); token budget; warmup fraction (secondary).

F. Failure modes / confounds
Declaring a schedule winner at a bad peak LR; unequal evaluation protocols; reading train loss as if it were val; stopping mid-cooldown inconsistently; promoting mid-run W&B as finals.

G. Cross-links & further tests
Reference run feeds P2(b) EMA/regret. Horizon-free alternatives in P2(d). Same bounce story explains P1(d) schedule effect on (B).

P2(b) — Eval-only EMA and regret

A. Experimental result — PENDING MEASUREMENT (still blank after Batch 2 **9/9**)

- TODO: raw vs EMA val for (\beta\in{0.99,0.999}); regret curves; residual schedule gaps after EMA.
- **PREDICTION** locked before Batch 2 completion (jobs log eval-only EMA). Do **not** treat schedule **MEASURED** finals as EMA results.

B. Landscape / geometry
[\bar\theta_0=\theta_0,\qquad \bar\theta_t=\beta_{\mathrm{EMA}}\bar\theta_{t-1}+(1-\beta_{\mathrm{EMA}})\theta_t.]
EMA averages along the bounced trajectory → closer to the river centerline in parameter space (distinct from P1's cross-mode (\theta_{\mathrm{avg}})).

C. Dynamics / optimizer interaction
EMA is **evaluation-only** (no training-state change); starter logs `val_ema{decay}_loss`. Longer timescale (\beta=0.999) is smoother but laggier than (0.99). Regret uses the P2(a) best raw final as reference unless stated otherwise.

D. Quantitative prediction template — PREDICTION (ledger `P2-ema-gap-frac`; B2-locked)

Quantity	PREDICTION band / ranking
Gap-closed fraction (f=1-G_{\mathrm{EMA}}/G_{\mathrm{raw}})	([0.5,0.75))
After best EMA, lowest schedules	`cos` or `wsd0.2` still favored
Absolute final val	not invented — relative gaps only

E. Counterfactual levers
EMA decays; which schedule is reference; when EMA is logged; peak LR retune before claiming gap closure.

F. Failure modes / confounds
Accidentally training with EMA weights; comparing EMA of one run to raw of another; including warmup-dominated early EMA in "final gap."

G. Cross-links & further tests
Function-space ensemble (P1) vs parameter EMA (P2). Dip/rewarm (P2c) tests whether "settling" is reversible. P3: lowering effective (\eta) during bounce changes (\eta\lambda).

P2(c) — Temporary LR dip then rewarm

A. Experimental result — **MEASURED** final (R) · mid-dip **PENDING** (Batch 2 job 4 finished)

- **MEASURED** (`media/p2/batch2_partial_finals.csv`; ledger `B2-P2-dip-R` / `P2-dip-final-R`):
  - Job 4 `a3-b2-p2-wsd0.2-dip` W&B `rms7iyyt`: final val (2.938987970352173).
  - Control job 3 `wsd0.2` `soosjcxr`: final val (2.93774151802063).
  - Final (R=L_{\mathrm{dip}}-L_{\mathrm{wsd0.2}} =0.0012464523315429688\in[-0.02,0.02)) — band **HIT**.
- Mid-dip (R_{\mathrm{mid}}) — **PENDING** / **not MEASURED** (no mid-run snapshot in partial CSV).
- **Oct 20 quiz implication:** final dip regret near zero matches the locked band — temporary dip does not hurt (or help) the final vs matched `wsd0.2` cooldown; do not invent a mid-dip score until a snapshot exists.

B. Landscape / geometry
Dip → less bounce → drop toward river; rewarm → bounce resumes. Final position may return near the control's river-wall orbit — transient gains need not survive.

C. Dynamics / optimizer interaction
`lr_dip=(start,bottom,end,scale)` multiplies any schedule; times as fractions of training; model/optimizer state **not** reset. Handout requires the same final cooldown on control and dipped run for fair finals.

D. Quantitative prediction template — PREDICTION vs MEASURED (final only)

Quantity	Tag	Band / value
Final (R=L_{\mathrm{dip}}-L_{\mathrm{ctrl}})	**MEASURED HIT**	(0.0012464523315429688\in[-0.02,0.02)) (PREDICTION band matched)
Mid-dip (R_{\mathrm{mid}})	**PENDING** / **PREDICTION** only	Band ((-\infty,-0.02)); Observed blank — not in partial CSV; **not MEASURED**
Larger/longer dips	**PREDICTION** (qualitative)	larger transient (

E. Counterfactual levers
Dip scale, duration, start time; peak LR of the parent schedule; cooldown length.

F. Failure modes / confounds
Missing final cooldown on control; resetting optimizer; measuring only train loss; starting dip before warmup finishes.

G. Cross-links & further tests
EoS (P3): dip changes (\eta) → expected (\lambda) retarget (cousin of P3(a)ii). Horizon-free schedules (P2d) avoid budget-specific cooldowns.

P2(d) — Horizon-free schedules (invsqrt, SGDR)

A. Experimental result — **MEASURED** (Batch 2 @ 9/9; invsqrt + SGDR finished; P3 sharpness also **MEASURED**)

- **MEASURED invsqrt** (`media/p2/batch2_partial_finals.csv`; ledger `B2-P2-invsqrt-dL`):
  - Job 5 `a3-b2-p2-invsqrt` W&B `1jri34em`: final val (3.020456075668335).
  - Tuned ref = best of `{linear, cos, wsd0.2}` = `linear` ((L^\star=2.9282028675079346)).
  - (\Delta L=L_{\mathrm{invsqrt}}-L_{\mathrm{linear}}=0.09225320816040039) vs PREDICTION ([0.01,0.08)) — band **MISS** (too high; in ([0.08,\infty))).
- **MEASURED SGDR** (`B2-P2-sgdr-dL` / job 6 W&B `51b3qq4n`): final val (3.0348048210144043).
  - (\Delta L=L_{\mathrm{sgdr}}-L_{\mathrm{linear}}=0.10660195350646973) — band **MISS** (in ([0.08,\infty))).
  - (\lvert L_{\mathrm{sgdr}}-L_{\mathrm{invsqrt}} \rvert=0.014348745346069336\in[0,0.03)) — closeness **HIT**.
- **Oct 20 quiz implication:** both HF (\Delta L) gaps were **larger** than `[0.01,0.08)`; SGDR stayed close to invsqrt. Keep direction "HF worse than budget-tuned."
- TODO still: multi-budget transfer without cooldown.

B. Landscape / geometry
Unknown horizon ⇒ cannot rely on a final "settle to river" phase sized to the budget; iterates may remain partially bouncing at stop → higher val than a budget-tuned cosine/`wsd0.2` with a long cooldown.

C. Dynamics / optimizer interaction
Invsqrt (Vaswani et al.): for (t\ge 1), [\alpha_t=\alpha_{\max}\min\bigl{t/w,\sqrt{w/t}\bigr}.]
SGDR: after warmup, cycle [\alpha_t=\alpha_{\min}+\tfrac12(\alpha_{\max}-\alpha_{\min}) \bigl[1+\cos(\pi s/T)\bigr];]
restart **LR only**, not model/optimizer. AdamW unchanged. Choose (w,T,\alpha_{\max}, \alpha_{\min}) independently of final budget. Starter: `warmup_steps=94`, `sgdr2000`, `min_lr_ratio=0.1`.

D. Quantitative prediction template — invsqrt + SGDR **MEASURED** (B2-locked `2026-10-11T00:32:07Z`)

Ledger ID	Batch 2 job	Quantity	PREDICTION band	Status
`B2-P2-invsqrt-dL`	`a3-b2-p2-invsqrt`	(\Delta L=L_{\mathrm{hf}}-L_{\mathrm{tuned}}) at (T_{\mathrm{long}})	([0.01,0.08))	**MEASURED** (0.09225320816040039) **MISS**
`B2-P2-sgdr-dL`	`a3-b2-p2-sgdr2000`	same (\Delta L)	([0.01,0.08))	**MEASURED** (0.10660195350646973) **MISS**
`B2-P2-sgdr-dL`	vs invsqrt	(\lvert L_{\mathrm{sgdr}}-L_{\mathrm{invsqrt}}\rvert)	([0,0.03))	**MEASURED** (0.014348745346069336) **HIT**
`P2-horizon-dL`	mid budget (not B2 stop)	usefulness	(\lvert\Delta L\rvert<0.02) competitive	**PREDICTION** only (full-stop HF both MISS)

Transfer test: freeze ((w,T)) across budgets — usefulness = unknown-horizon setting, not cherry-picked (T).

E. Counterfactual levers
(w), (T), (\alpha_{\max},\alpha_{\min}); stopping budgets; accidental cooldown (must avoid).

F. Failure modes / confounds
Accidentally adding cooldown; restarting optimizer state on SGDR; retuning (w,T) per budget (defeats the point); unfair peak-LR mismatch vs tuned baselines.

G. Cross-links & further tests
Practice-quiz style: transfer a ((w,T)) pair across budgets. Tie to P3: SGDR restarts briefly change (\eta\lambda^{\mathrm{pre}}). Tie to P2(c): both probe reversible vs permanent settling.

Problem 3 — The Edge of Stability

Quadratic GD: stable for (\eta\lambda_{\max}<2). Cohen et al.: full-batch GD progressively sharpens until (\eta\lambda_{\max}\approx 2) and rides the edge. Adam (\beta_1=0.9) frozen-preconditioner story: edge at (\eta\lambda^{\mathrm{pre}}_{\max}\approx 38).

Label rule: P3 SGD clean-edge (\lambda) and AdamW post-warmup (\eta\lambda^{\mathrm{pre}}) are **MEASURED HIT**; P3(a)ii η-change arms stay **PREDICTION** / **PENDING**.

P3(a)i — Full-batch clean case ((N=32), (\eta=0.01))

A. Experimental result — **MEASURED** (Batch 2 job `a3-b2-p3-0-sgd` / W&B `0dewkykj`; ledger `B2-P3-sgd-lam` / `P3-clean-edge` **HIT**)

- Median (\lambda_{\max}) over updates 1200–1500: **245.16314737956807** (n=12; window (\eta\lambda) median (2.451631473795681); `media/p2/p3_sgd_sharpness_summary.json`).

- PREDICTION band ([150,250)) → **HIT**; near (2/\eta=200); late train loss ↓ (final train (1.268615484237671) @ step 1499; `internal/batch2-p3-sgd-lam-score-status.md`).
- TODO still: publish (\lambda_{\max}) vs update / vs train-loss plots with dashed threshold at 200.

B. Landscape / geometry
Sharpness (\lambda_{\max}=\lambda_{\max}(H(\theta))) on a fixed probe. Progressive sharpening: curvature rises until the optimizer's stability wall. Riding the edge: loss can still decrease while (\lambda_{\max}) hovers near (2/\eta).

C. Dynamics / optimizer interaction
`full_batch(32, 1500, learning_rate=0.01)`: GD on first (N) sequences, 16× embedding init, 1500 updates. Not the default AdamW minibatch recipe — this is the Cohen-style clean case.

D. Quantitative prediction template — MEASURED (locked PREDICT `2026-10-11T00:32:07Z`; ledger `B2-P3-sgd-lam` / `P3-clean-edge` **HIT**)

Quantity	Target / mechanism	PREDICTION band	Observed
Median (\lambda_{\max}) updates 1200–1500	edge (2/\eta=200) at (\eta=0.01)	([150,250))	**245.16314737956807** (**HIT**)
Progressive sharpening then hover	Cohen EoS	direction: rise → ride near 200	window (\eta\lambda) median (2.451631473795681); summarize plateau (2.547386438768327)
Late train loss while at edge	Cohen-style	continues to decrease overall	final train (1.268615484237671) @ step 1499 (↓)
Batch 2 job	`a3-b2-p3-0-sgd`	—	`0dewkykj` finished

Source: `media/p2/p3_sgd_sharpness_summary.json` / `internal/batch2-p3-sgd-lam-score-status.md` / scorecard `B2-P3-sgd-lam`.

E. Counterfactual levers
(N), (\eta), init scale (16× embedding init per handout), update count.

F. Failure modes / confounds
Probe ≠ train sequences; too few Lanczos iterations; mistaking transient spikes for the edge; plotting during warmup as if at edge; promoting mid-run sharpness as final median.

G. Cross-links & further tests
Mid-run (\eta) changes (a-ii); minibatch Adam (b). River-valley bounce is the macroscopic cousin of edge-riding oscillations.

P3(a)ii — Change (\eta) mid-run (×0.5 / ×2 at 750)

A. Experimental result — **PENDING MEASUREMENT** / not started

- TODO: (\lambda_{\max}) trajectories after half / double; symmetry check (time-to-new-edge, overshoot).
- **Not in Batch 2 required 9-job manifest** (expand starter / later batch). Intervals below are **PREDICTION** only — **not MEASURED**.

B. Landscape / geometry
Edge location is (2/\eta). Halving (\eta) raises the allowed sharpness ceiling; doubling lowers it. Response need **not** be symmetric in time or overshoot (handout asks).

C. Dynamics / optimizer interaction
`lr_schedule="step750x0.5"` vs double at update 750. After (\eta\leftarrow\eta/2): expect further sharpening toward new edge. After (\eta\leftarrow 2\eta): expect (\lambda_{\max}) to drop or become unstable until below new edge.

D. Quantitative prediction template — **PREDICTION** (ledgers `P3-eta-half`, `P3-eta-double`; not B2 required)

Intervention	Target (2/\eta)	PREDICTION median (\lambda_{\max}) (1200–1500)
(\eta\leftarrow 0.005) at 750	400	([350,450])
(\eta\leftarrow 0.02) at 750	100	([80,120])
Time-to-retarget symmetry	—	**not** assumed equal

E. Counterfactual levers
Step location (750); factor (0.5 vs 2); (N); whether first edge was reached before the step.

F. Failure modes / confounds
Changing (\eta) before the first edge is reached; comparing absolute (\lambda) without (\eta\lambda); insufficient steps after the change to retarget.

G. Cross-links & further tests
P2 LR dip is the noisy minibatch cousin of this intervention. P4: (\lambda_{\max}) moves can also come from reparameterization without dynamics (invariance).

P3(b) — Minibatch AdamW preconditioned sharpness

A. Experimental result — **MEASURED** (Batch 2 job `a3-b2-p3-1-adamw` / W&B `foew12y0`; ledger `B2-P3-adamw-pre` / `P3-mb-edge` **HIT**)

- Post-warmup median (\eta\lambda^{\mathrm{pre}}): **30.987715914514** (n=25 post-warmup; primary scoring metric; `media/p2/p3_adamw_sharpness_summary.json`).

- PREDICTION band ([30,50)) → **HIT**; near Adam threshold 38. Secondary (not scoring): all-measured median (31.813196147279584); 100–180M-tok window median (22.496516434600995).
- TODO still: publish (\eta\lambda^{\mathrm{pre}}) vs tokens plot with dashed threshold at 38.

B. Landscape / geometry
[\lambda^{\mathrm{pre}}{*max*}=*lambda*{\max}(D^{-1/2}HD^{-1/2}),\quad D=\sqrt{\hat v}+\epsilon.]
Preconditioned sharpness is the relevant stability quantity for Adam-style steps.

C. Dynamics / optimizer interaction
`minibatch()`: constant LR, shorter horizon (compute). For Adam with (\beta_1=0.9) and frozen-preconditioner story: [\eta\lambda^{\mathrm{pre}}_{\max}\approx 2(1+\beta_1)/(1-\beta_1)=38.]
Minibatch noise may blur the edge vs full-batch GD's tight ride at 2.

D. Quantitative prediction template — MEASURED (ledger `B2-P3-adamw-pre` / `P3-mb-edge` **HIT**)

Quantity	PREDICTION band / ranking	Observed
Post-warmup median (\eta_t\lambda^{\mathrm{pre}}_{\max})	([30,50)) (near 38)	**30.987715914514** (**HIT**)
Secondary medians (not primary score)	all-measured / 100–180M tok	(31.813196147279584) / (22.496516434600995)
Vs full-batch GD (\eta\lambda\approx 2) ride	noisier / wider band — not stuck near 2, not (\to\infty)	post-warmup (\eta\lambda^{\mathrm{pre}}\approx 31) near 38; not at 2
Batch 2 job	`a3-b2-p3-1-adamw`	`foew12y0` finished

Source: `media/p2/p3_adamw_sharpness_summary.json` / `internal/batch2-p3-adamw-score-status.md` / scorecard `B2-P3-adamw-pre`.

E. Counterfactual levers
Batch size, (\beta_1), LR schedule (constant in starter recipe), horizon length.

F. Failure modes / confounds
Including warmup in edge plots; reading raw (\lambda_{\max}) instead of preconditioned; short-horizon artifacts; wrong (\beta_1) in the threshold formula.

G. Cross-links & further tests
P3(c) hparam exploration; P4 spectrum explains *which* directions dominate (\lambda_{\max}). P2 schedules modulate (\eta_t) hence the instantaneous edge.

P3(c) — Open-ended hyperparameter exploration (~18 runs)

A. Experimental result — **PENDING MEASUREMENT** / not started

- TODO: factors that help enter/stay at the edge; plots with threshold lines; ≤18 runs budget honesty.
- Example axis templates in §D are **PREDICTION** sketches only — zero MEASURED P3(c) outcomes.

B. Landscape / geometry
Anything that changes effective curvature or step size relative to curvature moves time-to-edge and stability. Goal: factors affecting tendency to **enter and stay** at the edge (handout).

C. Dynamics / optimizer interaction
Candidate axes (one knob per cluster): batch size, peak LR, (\beta_1/\beta_2), weight decay, schedule, init scale, probe size. Prefer axes that change (\eta\lambda^{\mathrm{pre}}) interpretable against threshold 38.

D. Quantitative prediction template — **PREDICTION** (pre-register ≤3 axes in ledger before launch)
Example templates (do not treat as measured):

- Batch↑ at fixed peak LR: **PREDICTION** — harder to stay tightly at edge (noisier (\eta\lambda^{\mathrm{pre}}); median band widens).
- Peak LR↑: **PREDICTION** — faster time-to-edge; risk of instability overshoot.
- (\beta_1)↑: threshold (2(1+\beta_1)/(1-\beta_1)) moves — compare medians to the **new** threshold, not blindly to 38.

E. Counterfactual levers
One knob per run cluster; keep compute budget honest (~18 runs).

F. Failure modes / confounds
Multiple simultaneous changes; under-tuning LR so “no edge” is just “too small (\eta)”; claiming edge absence without plotting the correct threshold.

G. Cross-links & further tests
Feed P4(b) interventions that move (\lambda_{\max}) without hurting loss. Report should separate “dynamics edge” from “parameterization sharpness” (P4b).

Problem 4 — Inside the Hessian

Objects: sharpness, normalized sharpness (\mathrm{tr}(H^2)/d), negative eigenvalues, outliers, block structure (family & layer). Tools: Lanczos HVPs, Hutchinson; probe = 64 sequences; FP32, TF32 off.

Label rule: AdamW Hessian steps **0 / 3 / 15 / 61 / 153 / 305 / 763 / 1526 / 3052 / 4578 / 6104 / 7630 / 9375** are **MEASURED** from `media/p4/stage-a/` JSON (+ summaries). Stage a

13/13 DONE. Handout final bands (\lambda_{\max}\in[4.5,6.0]), val (\in[2.925,2.931]) are **assignment-given**. Rebound note: `internal/p4-lambda-rebound-13of13.md`. Ledger HIT/MISS copied from `internal/p4-score-dryrun.md` (operator; `--apply` does not mutate docs) → `internal/p4-9375-hit-miss-summary.md`.

Trajectory (MEASURED 13/13). Collapse by 15 → dip (\sim 2.195) @1526 (earlyfit **HIT**) → rebound to (\lambda@9375=5.967310256347902) (**inside** handout `[4.5,6.0]`). Norm **HIT**; block-energy **HIT** (excess_final=0.08727119221832025). Rebound: `internal/p4-lambda-rebound-13of13.md`.

MISS mechanisms (predictive — PREDICT bands locked; scored vs MEASURED). From preregister (`docs/a3-p4-preregister-draft.md`) + file-exact `internal/p4-9375-hit-miss-summary.md`:

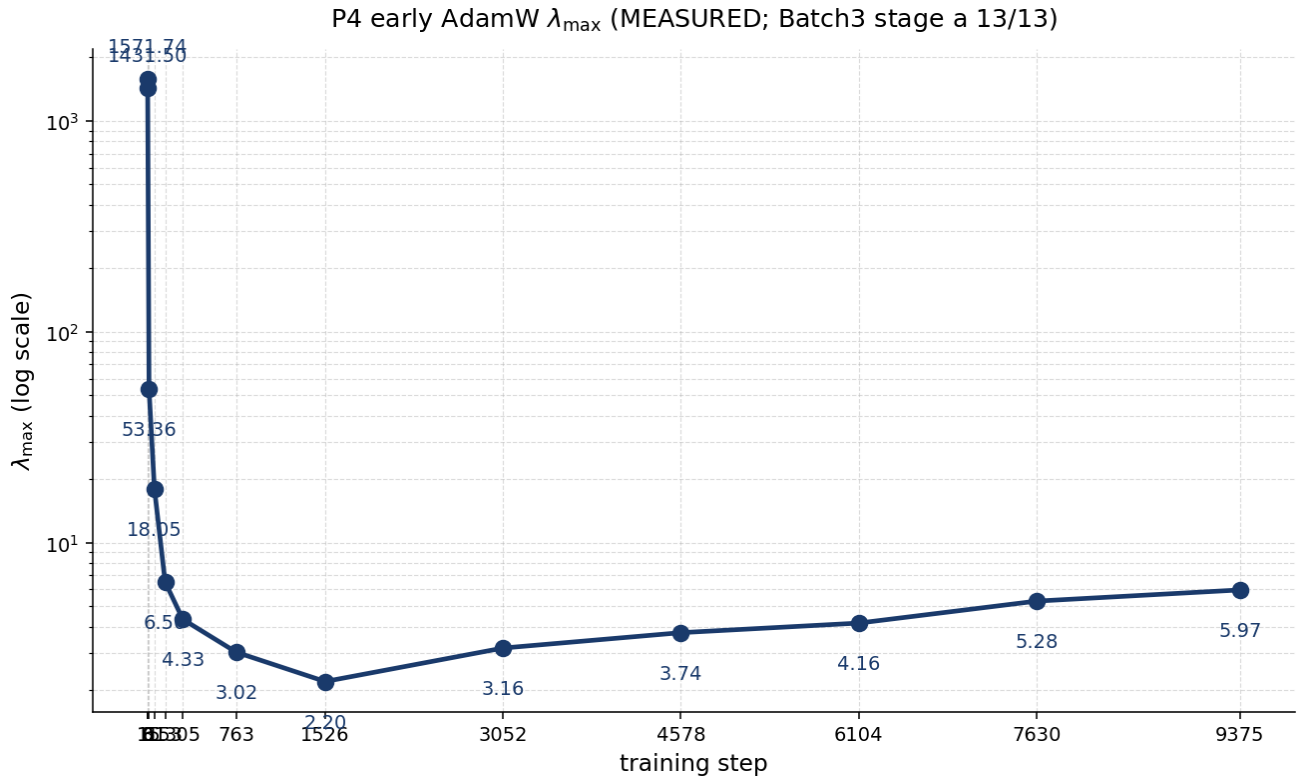
- **`P4-a-family-sharpest`:** PREDICT late ∈ {`attention`,`mlp`} (progressive sharpening in large residual blocks; embed/head early). MEASURED late=`head` ((5.641603135007765)); early_top2=`embedding`,`attention`; ratio (7.38436579244244\notin[1.5,5)). Residual-block late locus **failed**.
- **`P4-a-density-neg`:** PREDICT neg mass shrinks 0→9375; final ∈`[0,0.15)`. MEASURED (0.004510114309467541\to0.01491374251600709\to0.9344596369432798\to0.06965100420936976); final band OK, shrink_0→9375=**false**. Saddle-cleanup direction **failed**.
- **`P4-lam-9375-earlyfit`:** PREDICT `[0.3,4.0)` (early decay / preferred asymptote (\sim 3.51)). MEASURED (5.967310256347902) — **MISS** the earlyfit envelope, **HIT** the handout edge band `[4.5,6.0]` by rebound.

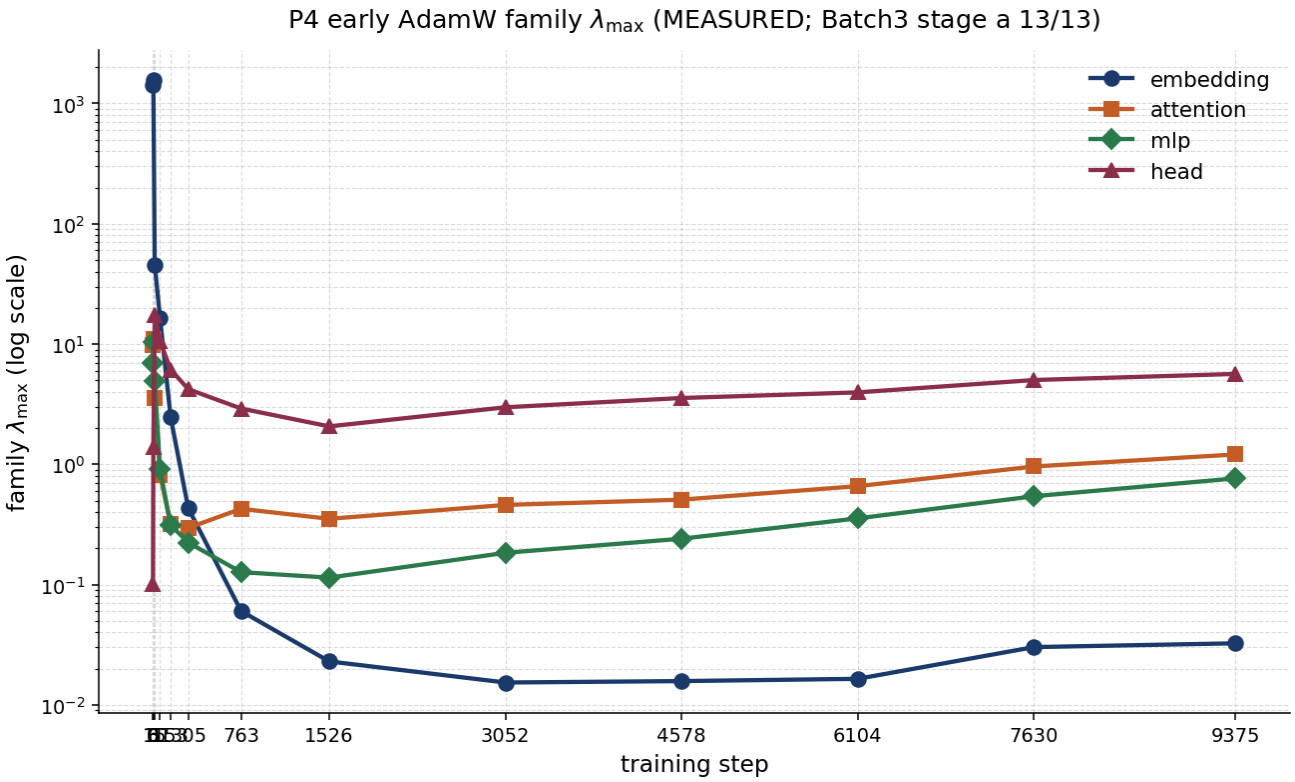
P4(a) — What the Hessian looks like over training (0 train runs; measure ckpts)

A. Experimental result — MEASURED (steps 0, 3, 15, 61, 153, 305, 763, 1526, 3052, 4578, 6104, 7630, 9375; stage a **13/13 DONE**)

- **MEASURED (\lambda_{\max}) trajectory** (summaries / stage-a JSON):

Step	Source	(\lambda_{\max}) (exact)
0	media/p4/a-adamw-0.summary.json	(1431.5030691763686)
3	media/p4/a-adamw-3.summary.json	(1571.7420262270407)
15	media/p4/a-adamw-15.summary.json	(53.36426571742344)
61	media/p4/a-adamw-61.summary.json	(18.054717868295125)
153	media/p4/stage-a/a-adamw-153.json	(6.502395300121604)
305	media/p4/stage-a/a-adamw-305.json	(4.32926630998023)
763	media/p4/a-adamw-763.summary.json	(3.02398152405625)
1526	media/p4/a-adamw-1526.summary.json	(2.1954286466134665)
3052	media/p4/a-adamw-3052.summary.json	(3.1637529312879873)
4578	media/p4/a-adamw-4578.summary.json	(3.7423705126158477)
6104	media/p4/a-adamw-6104.summary.json	(4.155164815834522)
7630	media/p4/a-adamw-7630.summary.json	(5.281516516770698)
9375	media/p4/a-adamw-9375.summary.json	(5.967310256347902)





- **Pattern (MEASURED 13/13):** collapse by 15 → minimum @1526 ((2.195)) → rebound to (\lambda@9375=5.967310256347902) (**inside** handout). Late family top = **head** ((5.642)) then attention ((1.208)). `density_runs` at **0, 153, 1526, 9375**.

- **FIT context (not MEASURED):** preferred early-fit asymptote (\sim 3.51) from `internal/p4-lambda-early-fit.md` — **fit / PREDICTION only**, never treat as an Observed (\lambda).

- **Early-fit vs MEASURED** (match `internal/p4-lambda-early-fit.md`; stage a **13/13**):

  - **Preferred** FITTED exp→asymptote on (s\ge 61): point PREDICT (\approx\mathbf{3.51}) at 1526 and 9375 — **FIT only**; **undershoots** late MEASURED.
  - MEASURED 1526 = (2.1954286466134665) (**HIT** in `[1.4,4.0)`); preferred ≈3.51 note-only miss. Rebound continues through **7630** ((5.2815)) → **9375** ((\mathbf{5.9673})) — still **above** ≈3.51 and **inside** handout `[4.5,6.0]`.
  - **Bands:** `P4-lam-1526-earlyfit` **`[1.4,4.0)` HIT**; `P4-lam-9375-earlyfit` **`[0.3,4.0)` MISS** (Observed (5.967310256347902)).
  - Aggressive power-only points **≈1.43** / **≈0.36** were **band floors only** — not the preferred PREDICT.
  - Fit asymptote (\sim 3.51) **undershoots** late MEASURED through **9375**. Dens rose mid-train then fell to (0.06965100420936976) @9375 — shrink vs init fails → dens-neg **MISS**.

- **Scored (ledger):** `P4-lam-1526-earlyfit` **HIT**; `P4-lam-9375-earlyfit` **MISS** ((5.967310256347902)); `P4-a-norm-sharpness` **HIT**; `P4-a-family-sharpest` **MISS**

(late=head); P4-a-density-neg **MISS**; P4-a-block-energy **HIT** (excess_final=0.08727119221832025; fam_above=True; layer_above=True).

- **MEASURED density:** neg_frac 0→153→1526→9375 = (0.004510114309467541\to 0.01491374251600709\to 0.9344596369432798\to 0.06965100420936976) — final in [0,0.15) but **no shrink** vs init → **MISS**.

- **Counterfactual prediction lessons (predictive MISSes; bands unchanged):**

  1. **Earlyfit @1526 HIT** — late rebound breaks the early-decay PREDICT: (\lambda@9375=5.967310256347902\notin[0.3,4.0)) yet lands in handout [4.5,6.0] (P4-lam-9375-earlyfit **MISS**).
  2. **Family PREDICT attn|mlp late** fails: sharpest=head ((5.641603135007765)); ratio (7.38436579244244\notin[1.5,5)).
  3. **Dens-neg PREDICT "shrink 0→9375"** fails: final (0.06965100420936976>) init (0.004510114309467541) (final still ∈[0,0.15)).
  4. Norm mid→final nondecreasing + handout final λ → **HIT** (separate ID).

- **Scored:** block-energy MEASURED **HIT** — excess_final=0.08727119221832025; fam_above=True; layer_above=True (a-random-energy-1526/9375).

B. Landscape / geometry
Families: embed, attn projections, QK-norm gains, MLP, other RMSNorm gains, output head. Off-diagonal blocks = dependencies between variables. If within-block Frobenius energy ≫ between-block (and above random partitions of same sizes), curvature is meaningfully block-diagonal.
MEASURED geometry lessons (13/13):

- Curvature collapses from init, **minimum** @1526, then **rebounds** through **9375** ((\lambda=5.967310256347902)). Late block (\lambda_{\max}) dominated by **head**. Fit asymptote (\sim 3.51) **undershoots** (FIT only).
- Dens rose to ~0.93 @1526 then (0.06965) @9375 — no shrink vs init (**MISS**). Random partition MEASURED — excess_final=0.08727119221832025 (**HIT**).

C. Dynamics / optimizer interaction
Provided AdamW checkpoints at 13 steps from 0→9375 (614M tokens; step 153≈10M, 1526≈100M). AdamW trajectory shapes which blocks sharpen. SGDM comparison ckpts provided for P4(c). See **Mechanism** above for the **13/13** (\lambda) minimum@1526 + rebound through **9375** ((5.967310256347902)).

D. Quantitative prediction template — PREDICT bands locked; MISSes scored file-exact

- **P4-lam-* earlyfit:** bands [1.4,4.0) / [0.3,4.0) (preferred FIT ≈3.51 note-only). @1526 **HIT**; @9375 **MISS** — MEASURED (5.967310256347902) rebounded into handout [4.5,6.0] (early-decay mechanism under-predicted late edge).
- **P4-a-family-sharpest:** PREDICT late attn|mlp + ratio [1.5,5) — **MISS** (late=head (5.641603135007765); ratio (7.38436579244244)). Early embed/head-among-top-2 story partly holds (early_top2=embedding,attention).

- **`P4-a-density-neg`:** PREDICT shrink 0→9375 + final `[0,0.15)` — **MISS** on shrink (final (0.06965100420936976>) init); final band OK.
- **`P4-a-block-energy`:** PREDICT family within > random, excess `[0.05,0.30)` — **HIT** (excess_final=0.08727119221832025; fam_above=True; layer_above=True).
- Handout finals (**assignment-given**): (\lambda_{\max}\in[4.5,6.0]), val (\in[2.925,2.931]).

E. Counterfactual levers
Checkpoint step choice; `density_runs`; family vs layer partition; extremes vs density-only jobs.

F. Failure modes / confounds
TF32 on; incomplete probe (<64 rows); comparing AdamW vs SGDM without noting optimizer; under-iterated Lanczos residuals.

G. Cross-links & further tests
Sets baseline for P4(b) loss-band interventions. Top eigenspace feeds P4(c). Edge (P3) is about (\lambda_{\max}) dynamics; here about full spectrum / blocks.

P4(b) — Move (\lambda_{\max}) without moving the loss

A. Experimental result — MEASURED (`P4-invar-dL` MISS)

- Before/after from dry-run `internal/p4-b-score-dryrun.md`: λ 5.967310256347902→2.780040977639008; |ΔL|=2.73853462934494; ratio=0.4658784038724411 → **MISS**.
- Paths: `/cursor/stores/bc-01947f99-cdd8-4c04-97a8-01a1028bf8af/media/p4/b-rescale-reference.json`, `/cursor/stores/bc-01947f99-cdd8-4c04-97a8-01a1028bf8af/media/p4/b-rescale-head-x0.25.json`.
- Training interventions (`P4-b-train-2x`) still **PENDING** (untouched).

B. Landscape / geometry
Loss invariances (lecture / handout hint): reparameterizations that preserve predictions can rescale curvature of affected blocks — so (\lambda_{\max}) is **not** a pure quality metric across runs or parameterizations.

C. Dynamics / optimizer interaction
Two routes: (1) hand transform on a final checkpoint (no further training); (2) training-recipe change that induces sharpening "for free." Hyperparams may get partway; code-level interventions may be stronger (handout).

D. Quantitative prediction template — PREDICTION (ledger `P4-invar-dL`; Observed PENDING) + MEASURED MISS
Hand transform on final ckpt (before = MEASURED reference above):

- **PREDICTION:** (\lvert\Delta L_{\mathrm{val}}\rvert\in[0,0.002)) (scorer uses file-exact (\lvert\Delta\mathrm{probe_loss}\rvert) until val lands).
- **PREDICTION:** (\lambda_{\max}^{\mathrm{after}}/\lambda_{\max}^{\mathrm{before}} \in[2,\infty)) **or** ((0,0.5)) (direction = rescale choice; (\ge 2\times) move).

- **MEASURED:** |ΔL|=2.73853462934494; λ ratio=0.4658784038724411; λ 5.967310256347902→2.780040977639008; probe_loss 2.9474442899227142→5.685978919267654 → **MISS** (from `internal/p4-b-score-dryrun.md`). Training route: final (\lambda_{\max}) outside ([4.5,6.0]) by (\ge 2\times) factor vs baseline mean, loss inside ([2.925,2.931]) — pre-register which knob before launch (`P4-b-train-2x`).

E. Counterfactual levers
Which family to rescale; WD / init / LR as soft sharpeners; optional small code change; seed (band is across three seeds).

F. Failure modes / confounds
Moving loss out of band; changing evaluation probe; claiming (\lambda_{\max}) comparable across inequivalent parameterizations; mistaking a failed invariance (predictions changed) for a curvature result.

G. Cross-links & further tests
Never treat raw (\lambda_{\max}) as comparable without stating parameterization. P3 edge thresholds use (\eta\lambda) / (\eta\lambda^{\mathrm{pre}}) — still parameterization-sensitive via (H). Quiz implication: sharpness ≠ quality.

P4(c) — Training in / out of the top subspace (SGDM)

A. Experimental result — MEASURED

- From `internal/p4-c-score-dryrun.md` / `media/p4/stage-c/`: mean top_frac=0.09296940652964016 → **MISS**; L_full=3.582527816295624, L_top=5.502330541610718, L_removed=3.5821831226348877, gap=-0.000344693660736084 → **MISS**.
- Optional `P4-c-k-ablate` / stale-evecs still **PENDING** (untouched).

B. Landscape / geometry
Gur-Ari et al.: gradient lies mostly in span of top few Hessian eigenvectors. Top subspace ≈ directions of highest curvature / progress. Removing that span should starve updates of useful progress if the claim holds.

C. Dynamics / optimizer interaction
SGD+momentum easier to reason about than Adam. From SGDM ckpt @ 100M tokens: **zero momentum**, continue 300 steps with transform ∈ {`full`,`top`,`removed`}; refresh eigenvectors every 25 steps. Log losses, (|P_{10}g|/|g|), (\lambda_{\max}) vs (2(1+\beta)/\eta_t).

D. Quantitative prediction template — PREDICTION (ledger `P4-top10-frac`) + **MEASURED** + **MEASURED**

- **PREDICTION:** mean (|P_{10}g|/|g|\in[0.6,0.85)) over the continue.
- **PREDICTION (ranking):** full (\approx) top-10 (<) removed (removed worst).
- **MEASURED (`P4-top10-frac`):** mean=0.09296940652964016 → **MISS**.

- **MEASURED (`P4-c-loss-rank`):** L_full=3.582527816295624; L_top=5.502330541610718; L_removed=3.5821831226348877; gap=-0.000344693660736084 → **MISS**.
- If surprises: vary (k), refresh, LR, start point (handout warns results may be unexpected).

E. Counterfactual levers
(k), refresh interval, (\eta), start checkpoint, (\beta).

F. Failure modes / confounds
Stale eigenvectors; Adam instead of SGDM; forgetting to zero momentum at continue; comparing to AdamW (\lambda_{\max}) bands without care.

G. Cross-links & further tests
River-valley: top directions may be the "wall-normal" bounce axes. Optional P5 quiz from an unexpected subspace result. Edge (P3): heavy-ball threshold line is the SGDM cousin of (\eta\lambda=2).

Problem 5 — Optional own prediction problem

Label rule: P5 is **MEASURED** (6/7) from `media/p5/` / `internal/p5-score-dryrun.md`. `P5-B-cool-x0` **MISS**; `P5-dB-cool` **MISS**; `P5-B-const-ctrl` **HIT**; `P5-rank-arms` **MISS**; `P5-G-cool-fitC` **HIT**; `P5-emergent-basin` **MISS**; `P5-L-cool-dir` **INSUFFICIENT** (`dL=null` / (L_R,L_C) missing — no HIT/MISS).

MISS mechanisms (predictive — PREDICT bands locked; scored vs MEASURED). From `media/p5/` / `internal/p5-score-dryrun.md`:

- **Cooldown did not collapse barrier:** `P5-B-cool-x0` / `P5-dB-cool` / `P5-rank-arms` all **MISS** — B_C=0.7697049379348755 stays ≫0.01; dB=-0.14600493793487546 (negative vs PREDICT `[0.5,0.8)`); ranking (B_C < B_R) and (B_C < B_K) fails.
- **Constant-LR control stayed high-(B):** `P5-B-const-ctrl` **HIT** (B_K=0.7320129871368408 ∈ `[0.5,∞)`).
- **Fit C held:** `P5-G-cool-fitC` **HIT** (|G−(B+0.09)|=0.020580991902542256 < 0.05).
- **`P5-emergent-basin`:** PREDICT same-valley walls → MEASURED not-same-valley / JSON emergent='true-modes' (**MISS**).
- **`P5-L-cool-dir`: INSUFFICIENT** (dL=null) — no direction takeaway.

P5(a–b) — Question writeup + compact resolving code

A. Experimental result — MEASURED

- File-exact: B_R=0.6237, B_C=0.7697049379348755, B_K=0.7320129871368408, G_C=0.8391239460323332, dB=-0.14600493793487546; dL=null.
- Scores: `P5-B-cool-x0` **MISS**; `P5-dB-cool` **MISS**; `P5-B-const-ctrl` **HIT**; `P5-rank-arms` **MISS**; `P5-G-cool-fitC` **HIT**; `P5-emergent-basin` **MISS**; `P5-L-cool-dir` **INSUFFICIENT** (dL / (L_R, L_C) missing).

- Takeaway: cooldown did not collapse the barrier; constant-LR control stayed high-(B); Fit C held; emergent='true-modes' (not same-valley).
- Source: `internal/p5-score-dryrun.md` / `media/p5/`.

B. Landscape / geometry
Choose a phenomenon already seen in P1–P4 (barrier, EMA gap, edge retarget, invariance, subspace). Prefer something where the linear path / threshold / ranking is unambiguous.

C. Dynamics / optimizer interaction
Prefer a single clear intervention with a stable predicted interval. Graded runs: seed 42; executable from starter + small diff (handout).

D. Quantitative prediction template — PREDICTION (author before coding) + **MEASURED**

- **MEASURED (`P5-B-cool-x0`):** B_C=0.7697049379348755 → **MISS** (cooldown did not collapse barrier to (\le 0.01)).
- **MEASURED (`P5-dB-cool`):** dB=-0.14600493793487546 (B_R=0.6237 [JSON] − B_C=0.7697049379348755) → **MISS** (negative; no large collapse vs PREDICT `[0.5,0.8)`).
- **MEASURED (`P5-B-const-ctrl`):** B_K=0.7320129871368408 → **HIT** (constant-LR control stayed high-(B)).
- **MEASURED (`P5-rank-arms`):** B_C=0.7697049379348755 B_R=0.6237 B_K=0.7320129871368408 → **MISS** (cooldown arm not below raw or control).
- **MEASURED (`P5-G-cool-fitC`):** G_C=0.8391239460323332 B_C=0.7697049379348755 |G-(B+0.09)|=0.020580991902542256 → **HIT** (Fit C held).
- **MEASURED (`P5-emergent-basin`):** B_C=0.7697049379348755 B_K=0.7320129871368408 → not-same-valley (JSON emergent='true-modes') → **MISS**.
- **INSUFFICIENT (`P5-L-cool-dir`):** dL / (L_R, L_C) missing (`media/p5/` dL=null) — no HIT/MISS. Mirror A2 quiz structure:

1. “From the assignment” (reference cell).
2. “This question changes” (one knob).
3. Interval / ranking / threshold options with adjacent-half scoring.
   Pre-register the answer interval in the ledger before running the resolving script.

E. Counterfactual levers
Keep scope to <100 LOC; one seed; one metric printed.

F. Failure modes / confounds
Ambiguous grading; needing huge sweeps; answer depending on flaky seeds; inventing “measured” answers in the writeup before the diff runs.

G. Cross-links & further tests
Staff may reuse; align style with `docs/a3-counterfactual-prediction-quiz.md`. Good seeds: P1 ens-vs-avg gap, P2 dip final (R), P3 η retarget asymmetry, P4 invariance (\Delta L).

Cross-problem prediction habits

1. Name the **reference cell** and the **single changed knob**.
2. Prefer **interval / ranking / threshold** answers tied to handout constants ((B\le 0.01), (\eta\lambda=2), (\eta\lambda^{\mathrm{pre}}=38)).
3. Tag every number: measured / fitted / extrapolated / **PREDICTION**.
4. Pre-register in `docs/a3-prediction-ledger.csv` (see `docs/a3-prediction-ledger.md`) before launch — **MD/CSV only, not a PDF**.
5. Never paste study-guide speculation into the official report as if measured.
6. Practice items: optional `docs/a3-counterfactual-prediction-quiz.md` (PDF #3 if exported).
7. Batch 1 measure landed: P1 §A and ledger Observed filled only from `media/p1/*` — do not backfill invented precision. Batch 1 **FITTED** curves live in Appendix Q1.1; out-of-grid rows are Kind=`extrapolated`.
8. Batch 2 PREDICTIONs locked `2026-10-11T00:32:07Z`. **MEASURED** @ **9/9**: schedule (R), dip **final** (R) (**HIT**), both HF (\Delta L) (**MISS**) + vs-invsqrt (**HIT**), P3 SGD (\lambda) (**HIT**), AdamW (\eta\lambda^{\mathrm{pre}}) (**HIT**). Batch 3: **P4(a)** stage-a **13/13** MEASURED + scored; **P5** MEASURED 6/7 (`P5-L-cool-dir` **INSUFFICIENT**). Still **PENDING**: **mid-dip** (R_{\mathrm{mid}}), EMA, P3 η-change, **P4(b)/(c)**.

Appendix Q — Quantitative prediction card

Equations and thresholds; compact MD twin: `docs/a3-quantitative-prediction-reference.md` (not a PDF). P1 empirical domain filled from MEASURED Batch 1. P2 + P3 starters **MEASURED** @ 9/9 (both sharpness HITs); **P4(a)** stage-a **13/13** MEASURED + scored; **P5** MEASURED 6/7 (`P5-L-cool-dir` **INSUFFICIENT**). Still **PENDING MEASUREMENT**: **mid-dip / EMA / P3 η-change / P4(b)/(c)**.

Q1. Mode connectivity (P1)

[B(\theta_1,\theta_2)

L!\left(\frac{\theta_1+\theta_2}{2}\right)

\frac{L(\theta_1)+L(\theta_2)}{2}]

Symbol / rule	Meaning
(L)	Validation loss
(B\le 0.01)	Weight averaging “works”

[\theta_{\mathrm{avg}}=\tfrac12\theta_1+\tfrac12\theta_2,\quad \theta(\alpha)=(1-\alpha)\theta_1+\alpha\theta_2,\ \alpha\in{0,0.25,0.5,0.75,1}]

[p_{\mathrm{ens}}(\cdot\mid c)=\tfrac12 p_{\theta_1}(\cdot\mid c)+\tfrac12 p_{\theta_2}(\cdot\mid c)]

Grid: (x\in{0,50,100,250,500,1000}), (y\in{25,50,100,250,500}), `wsd0.0`; fork copies model + optimizer state.

MEASURED domain (Batch 1):

- (26/30) cells with (B\le 0.01); failures ((0,250),(0,500),(50,500),(100,500)).
- (B\in[-0.1161,0.6237]), mean (-0.0334).
- On failure cells, (G=L(\theta_{\mathrm{avg}})-L_{\mathrm{ens}}\in[0.1311,0.7281]) (median (\approx 0.279)).
- Full tables: `docs/a3-experimental-appendix.md`, `media/p1/barrier-table.csv`.
- Plots (matplotlib): `media/p1/plots/B_heatmap.png`, `B_vs_y_curves.png`, `B_vs_x_y500.png`, `ensemble_vs_avg.png`.

Ledger: `P1-B-early-long`, `P1-ens-gap`, `P1-interp-peak`, `P1-grid` → Kind=`measured`, resolved; `P1-d-dB-cos` still **PREDICTION**; extrapolated rows `P1-B-x0-y750`, `P1-B-x25-y500`, `P1-B-x2000-y500`, `P1-G-x0-y750` (Kind=`extrapolated`).

Q1.1 FITTED relationships (Batch 1 MEASURED only)

No invented measurements. All coefficients from `media/p1/barrier-table.csv` / `barrier-summary.json`. Power laws not forced; preferred forms chosen by (R^2) and residual structure among simple candidates (linear / quadratic / (\log(1+x)) / (\exp)).

Fit A — (B) vs (y) at fixed (x=0)

Model	Domain	Equation	(R^2)	RMSE	Resid
Quadratic (preferred full curve)	(x=0), (y\in{25,50,100,250,500})	(B=-0.0765+7.61\times10^{-4},y+1.30\times10^{-6},y^2)	0.991	0.0243	(+0.0
Linear growth regime	(x=0), (y\ge 100) ((n=3))	(B=-0.1901+0.001628,y)	(\approx 1.000)	0.0003	(-0.00

Validity: growth-regime linear is almost exact on the three MEASURED points with (y\ge 100) but (n=3) — treat out-of-grid use as **extrapolated** with widening bands. For (y\in{25,50}), (B) is mildly negative (connected); do not apply the growth-regime line there.
Sibling slopes (linear (B=a+by), full (y) grid): (x=50): (R^2=0.989), slope (5.14\times10^{-4}); (x=100): (R^2=0.968), slope (2.83\times10^{-4}); (x=250): (R^2=0.833); (x\ge 500): (R^2\le 0.26) (flat — no useful (y)-trend).

Fit B — (B) vs (x) at fixed (y=500)

Model	Domain	Equation	(R^2)	RMSE
Exp decay (**preferred**)	(y=500), (x\in{0,50,100,250,500,1000})	(B=-0.0663+0.6870,e^{-0.02316,x})	0.995	0.0174
(\log(1+x))	same	(B=0.5850-0.1079\log(1+x))	0.964	0.0472
(a+b/\sqrt{x+1})	same	(B=-0.0651+0.7000/\sqrt{x+1})	0.960	—

Exp residuals (by (x)): (+0.003,-0.020,+0.030,+0.012,-0.010,-0.016).
Validity: preferred on MEASURED (x\le 1000) at (y=500). As (x\to\infty), Fit B asymptotes to (B\approx -0.066) (connected). At smaller (y), (\log(1+x)) fits weaken ((R^2=0.28) at (y=25); (0.95) at (y=250)) — barrier-vs-prefix structure is strongest at large (y).

Fit C — ensemble–avg gap (G) vs (B)

[G=L(\theta_{\mathrm{avg}})-L(p_{\mathrm{ens}}).]

Model	Domain	Equation	(R^2)	RMSE
Linear (all cells)	30 MEASURED cells	(G=0.0897+1.0328,B)	0.989	0.0151
Linear ((B>0.01))	4 failure cells	(G=0.1020+1.0066,B)	0.9997	0.0039

Identity check: (G=B+(\bar L-L_{\mathrm{ens}})) with (\bar L=\tfrac12(L_1+L_2)). MEASURED (\bar L-L_{\mathrm{ens}}) has mean (0.0886) (std (0.0158)), nearly independent of (B) ((R^2=0.086)) — so (G\approx B+0.09) is the mechanistic reading of Fit C.

Extrapolations beyond the MEASURED grid (PREDICTION / Kind=`extrapolated`)

Ledger ID	Cell	Point from fits	Interval (grounded)	Models
`P1-B-x0-y750`	((0,750))	quad (1.22); growth-lin (1.03)	(B\in[0.9,1.4))	Fit A
`P1-B-x25-y500`	((25,500))	exp (0.319); lin-interp MEASURED endpoints (0.377)	(B\in[0.25,0.40))	Fit B + MEASURED ((0,500),(50,500))
`P1-B-x2000-y500`	((2000,500))	exp asymptote (-0.066)	(B\in[-0.12,-0.01))	Fit B
`P1-G-x0-y750`	((0,750))	(G(B_{\mathrm{quad}})=1.35); (G(B_{\mathrm{lin}})=1.15)	(G\in[1.0,1.5))	Fit C ◦ Fit A

Not yet measured — do not treat as Observed.

Q2. River-valley regret and EMA (P2)

[R_M(t)=L(\theta^M_t)-L(\theta^{\mathrm{ref}}_t)]

[\bar\theta_0=\theta_0,\qquad \bar\theta_t=\beta_{\mathrm{EMA}}\bar\theta_{t-1}+(1-\beta_{\mathrm{EMA}})\theta_t]

Starter EMA decays: ({0.99,0.999}) (eval only).

Invsqrt: (\alpha_t=\alpha_{\max}\min{t/w,\sqrt{w/t}}) for (t\ge 1).
SGDR: (\alpha_t=\alpha_{\min}+\tfrac12(\alpha_{\max}-\alpha_{\min})[1+\cos(\pi s/T)]); restart LR only.
LR dip: `lr_dip=(start,bottom,end,scale)`; match control's final cooldown.

Tag	Domain @ Batch 2 finished 9/9 + P3 sharpness
MEASURED	Four schedules; dip **final** (R=0.0012464523315429688) (**HIT**); invsqrt (\Delta L=0.09225320816040039) (**MISS**); SGDR (\Delta L=0.10660195350646973) (**MISS**); (\lvert L_{\mathrm{sgdr}}-L_{\mathrm{inv}}\rvert=0.014348745346069336) (**HIT**); P3 SGD (\lambda=245.16314737956807) (**HIT**); AdamW (\eta\lambda^{\mathrm{pre}}=30.987715914514) (**HIT**)
PENDING	**Mid-dip** (R_{\mathrm{mid}}) (not logged); EMA gap-closed fraction; P3 η-change arms

Q2.1 Batch 2 PREDICTION table (locked `2026-10-11T00:32:07Z`)

PREDICTION bands locked before Batch 2 completion (mid-run W&B not used to tune). Manifest: 7 P2 + 2 P3 jobs. Status column separates resolved MEASURED from still-PENDING.

ID	Job	PREDICTION	Status @ 9/9
`B2-P2-wsd0.0-R`	`a3-b2-p2-wsd0.0`	Final (R\in[0.02,0.08)); worst of four	**MEASURED** (band MISS; worst HIT)
`B2-P2-linear-R`	`a3-b2-p2-linear`	Final (R\in[0,0.02)); (<L_{\mathrm{wsd0.0}})	**MEASURED** HIT
`B2-P2-cos-R`	`a3-b2-p2-cos`	Final (R\in[0,0.02)); top-2	**MEASURED** HIT
`B2-P2-wsd0.2-R`	`a3-b2-p2-wsd0.2`	Final (R\in[0,0.02)); top-2	**MEASURED** (band HIT; top-2 MISS)
`P2-ema-gap-frac`	jobs 1–4 EMA	(f\in[0.5,0.75))	**PENDING** (not MEASURED)
`B2-P2-dip-R`	dip job	Final (R\in[-0.02,0.02))	**MEASURED** HIT
`B2-P2-dip-R` (mid)	same run	mid-dip (R_{\mathrm{mid}}<-0.02)	**PENDING** (not MEASURED)
`B2-P2-invsqrt-dL`	`a3-b2-p2-invsqrt`	(\Delta L\in[0.01,0.08)) vs tuned	**MEASURED** MISS
`B2-P2-sgdr-dL`	`a3-b2-p2-sgdr2000`	(\Delta L\in[0.01,0.08)); (\lvert\Delta_{\mathrm{hf}}\rvert\in[0,0.03))	**MEASURED** (\Delta L) MISS; vs invsqrt HIT

PREDICTION pointers: `P2-sched-rank`, `P2-ema-gap-frac`, `P2-dip-final-R`, `P2-horizon-dL`, `B2-P2-*`. See Q2.1b for MEASURED schedule / dip / invsqrt / SGDR (@ 9/9).

Q2.1b Batch 2 MEASURED — schedules + dip/invsqrt/SGDR (P2 slice; full batch 9/9)

Jobs 0–6 finished (`media/p2/batch2_partial_finals.csv` exact vals). P3 sharpness also **MEASURED** (see Q3). **P4(a)** stage-a **13/13** MEASURED + scored (see Q4). **P5** MEASURED 6/7 (`P5-L-cool-dir` **INSUFFICIENT**). Still **PENDING**: **mid-dip / EMA / P4(b)/(c)**.
(R=L-L^\star) with (L^\star=L_{\mathrm{linear}}=2.9282028675079346).

Rank	Job	Label	W&B	Final train	Final val	(R)	vs PREDICTION
1	1	`a3-b2-p2-linear`	`r0294p93`	2.886288642883301	2.9282028675079346	0	band ([0,0.02)) **H** (<L_{\mathrm{w **HIT**
2	2	`a3-b2-p2-cos`	`sktw67q8`	2.8933908939361572	2.934119462966919	0.005916595458984375	band **HIT**; top-2
3	3	`a3-b2-p2-wsd0.2`	`soosjcxr`	2.8900909423828125	2.93774151802063	0.009538650512695312	band **HIT**; top-2 **MISS**
4	0	`a3-b2-p2-wsd0.0`	`widt420l`	3.0593721866607666	3.0975050926208496	0.16930222511291504	worst **HIT**; band ([0.02,0.08)) **MI**

Job	Label	W&B	Final val	Quantity	vs PREDICTION
4	`wsd0.2`+dip	`rms7iyyt`	2.938987970352173	(R_{\mathrm{dip}} =0.0012464523315429688) vs job3	([-0.02,0.02)) **HIT**
5	invsqrt	`1jri34em`	3.020456075668335	(\Delta L=0.09225320816040039) vs linear	([0.01,0.08)) **MISS**
6	SGDR (T=2000)	`51b3qq4n`	3.0348048210144043	(\Delta L=0.10660195350646973) vs linear	([0.01,0.08)) **MISS**
6	(vs invsqrt)	`51b3qq4n`	3.0348048210144043	(\lvert\Delta_{\mathrm{hf}} \rvert=0.014348745346069336)	([0,0.03)) **HIT**

`P2-sched-rank` HIT. Mid-dip / EMA still blank (**PENDING**). Both P3 starter (\lambda) / (\eta\lambda^{\mathrm{pre}}) **HIT**.

Q3. Edge of stability (P3)

Status: Batch 2 starter IDs **MEASURED HIT** — `B2-P3-sgd-lam` median (\lambda_{\max} =245.16314737956807); `B2-P3-adamw-pre` post-warmup median (\eta\lambda^{\mathrm{pre}} =30.987715914514). η-change arms (`P3-eta-half` / `P3-eta-double`) remain **PENDING** / **PREDICTION**.

[\lambda_{\max}=\lambda_{\max}\bigl(H(\theta)\bigr),\qquad \eta\lambda_{\max}<2\ \text{(GD/SGD quadratic edge)}]

[\lambda^{\mathrm{pre}}{*max*}=*lambda*{\max}(D^{-1/2}HD^{-1/2}),\quad D=\sqrt{\hat v}+\epsilon]

[\eta\lambda^{\mathrm{pre}}_{\max}<2(1+\beta_1)/(1-\beta_1)=38\quad(\beta_1=0.9,\ \text{frozen-precond story})]

Clean-case defaults	**Value**
(N), (\eta), updates	32, 0.01, 1500
Edge at (\eta=0.01)	(\lambda=200)
After (\eta\leftarrow 0.005) at 750	target (\lambda=400)
After (\eta\leftarrow 0.02) at 750	target (\lambda=100)

Domain: SGD clean-edge + AdamW post-warmup (\eta\lambda^{\mathrm{pre}}) = **MEASURED** (`media/p2/p3_*_sharpness_summary.json`). Half vs double asymmetry still **PENDING**.

Q3.1 Batch 2 PREDICTION table (locked `2026-10-11T00:32:07Z`)

ID	**Job**	**PREDICTION**	**Status**
`B2-P3-sgd-lam` / `P3-clean-edge`	`a3-b2-p3-0-sgd`	Median (\lambda_{\max}) (1200–1500) (\in[150,250)); late train loss ↓	**HIT** · (\lambda=245.16314737956807)
`B2-P3-adamw-pre` / `P3-mb-edge`	`a3-b2-p3-1-adamw`	Post-warmup median (\eta\lambda^{\mathrm{pre}}\in[30,50))	**HIT** · (30.987715914514)
`P3-eta-half`	*not in B2 required 9*	After (\eta\leftarrow 0.005): median (\lambda\in[350,450))	**PENDING** / **PREDICTION**
`P3-eta-double`	*not in B2 required 9*	After (\eta\leftarrow 0.02): median (\lambda\in[80,120))	**PENDING** / **PREDICTION**

Pointers: `P3-clean-edge` / `B2-P3-sgd-lam` and `P3-mb-edge` / `B2-P3-adamw-pre` both **MEASURED HIT**; η-change arms still PREDICTION / PENDING.

Q4. Hessian structure (P4)

Status: Stage a **13/13** MEASURED. Ledger (from dry-run DRAFT_*): `P4-a-norm-sharpness` **HIT**; `P4-a-family-sharpest` / `P4-a-density-neg` / `P4-lam-9375-earlyfit` **MISS**; `P4-lam-1526-earlyfit` **HIT**; `P4-a-block-energy` **HIT** (excess_final=0.08727119221832025; fam_above=True; layer_above=True). See `internal/p4-9375-hit-miss-summary.md`.

Predictive MISS notes (bands locked): late sharpest=`head` not attn|mlp; dens-neg no shrink 0→9375 (final (0.06965100420936976>) init); earlyfit@9375 (5.967310256347902) rebounded into handout `[4.5,6.0]` past `[0.3,4.0)`.

- Probe: 64 sequences; FP32, TF32 off; Lanczos + Hutchinson.

- Handout baseline band (**assignment-given**): (\lambda_{\max}\in[4.5,6.0]), val (\in[2.925,2.931]).
- **Early fit:** `P4-lam-1526-earlyfit` **HIT**; `P4-lam-9375-earlyfit` **MISS** ((5.967310256347902)).
- P4(b)/(c) scored: `P4-invar-dL` **MISS**. `P4-a-block-energy` **HIT** (excess_final=0.08727119221832025). Optional train/k-ablate still PENDING.
- Heavy-ball log line: (2(1+\beta)/\eta_t).
  Domain: resolvable `P4-a-*` + earlyfit scored; block-energy **HIT**; b waiting `internal/p4-b-waiting.md`.

PREDICTION pointers (still open): `P4-invar-dL` (PENDING after-arm). `P4-a-block-energy` scored **HIT**.

Q5. Quick threshold card

Handout / mechanism thresholds only (not Batch-2 MEASURED outcomes). P5 answer = **MEASURED** 6/7 (`P5-L-cool-dir` **INSUFFICIENT**).

Name	Value	Problem
Averaging works	(B\le 0.01)	P1
GD / SGD quadratic edge	(\eta\lambda_{\max}=2)	P3
Adam β1=0.9 precond edge	(\eta\lambda^{\mathrm{pre}}_{\max}=38)	P3
EMA decays (starter)	(0.99,\ 0.999)	P2
P4 loss band	val (\in[2.925,2.931])	P4
P4 sharpness band	(\lambda_{\max}\in[4.5,6.0])	P4
P4(b) move	(\lambda_{\max}) change (\ge 2\times)	P4

Q6. Ledger pointer (not part of this PDF's "results")

Machine-editable rows: `docs/a3-prediction-ledger.csv`. Human schema/notes: `docs/a3-prediction-ledger.md`. Fill **Observed** only from measurements.

- P1 resolvable (`P1-B-early-long`, `P1-ens-gap`, `P1-interp-peak`, `P1-grid`): Kind=`measured`, Status=`resolved`.
- Batch 2 scored (`B2-P2-*`, `B2-P3-sgd-lam`, `B2-P3-adamw-pre`, …): Observed filled only where finals exist — mark `resolved` + HIT/MISS.
- Fit-extrapolated (`P1-B-x0-y750`, …): Kind=`extrapolated`, Observed empty until a future measure.
- Still **PENDING** / Kind=`predicted`: EMA, mid-dip, P3 η-change, **`P4-invar-dL`** (after-arm), **P4(c)**. (**P4(a)** resolvable rows MEASURED/scored @ 13/13; P4(b) soft baseline MEASURED `scale={}`; `P4-a-block-energy` **HIT**; **P5** MEASURED 6/7; `P5-L-cool-dir` **INSUFFICIENT**.)